Question 1:

a) 4 points.

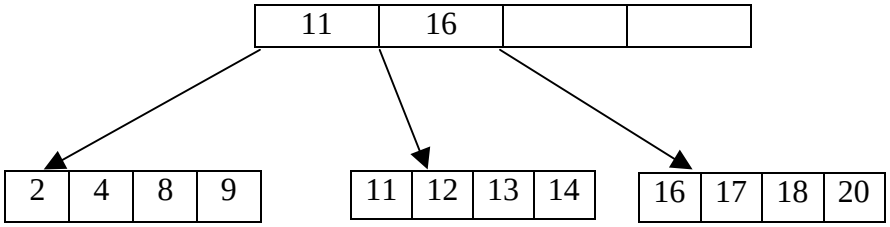
i. True, since the file is sorted on name

ii. True, you can build a dense index on any file

iii. False, the file is not sorted by phone

iv. False, the file is not primarily sorted by eno

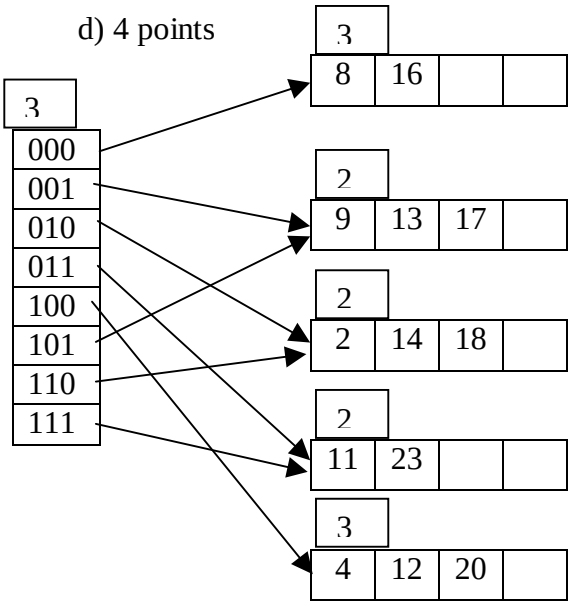
b) 2 points



c) 5 points

9 deletions

d) 4 points



e) 5 points

2 entries, Insertion of 2 entries into the second, third, or last will cause overflow, which will then cause a split.

f) 5 points

2 entries, Insertion of 2 entries into the last bucket will cause a split. Since the global == local depth, the global depth will increase by one.

Question 2:

a) 18 points
   i. E
   ii. E
   iii. F
   iv. F
   v. A
   vi. B

b) Full credit was given to everyone (3 points)

c) 4 points

$\Pi_{filename,\ song}$(User **NATURAL JOIN** Has **NATURAL JOIN** Song) / Π_{speed} (User)

d) 5 points

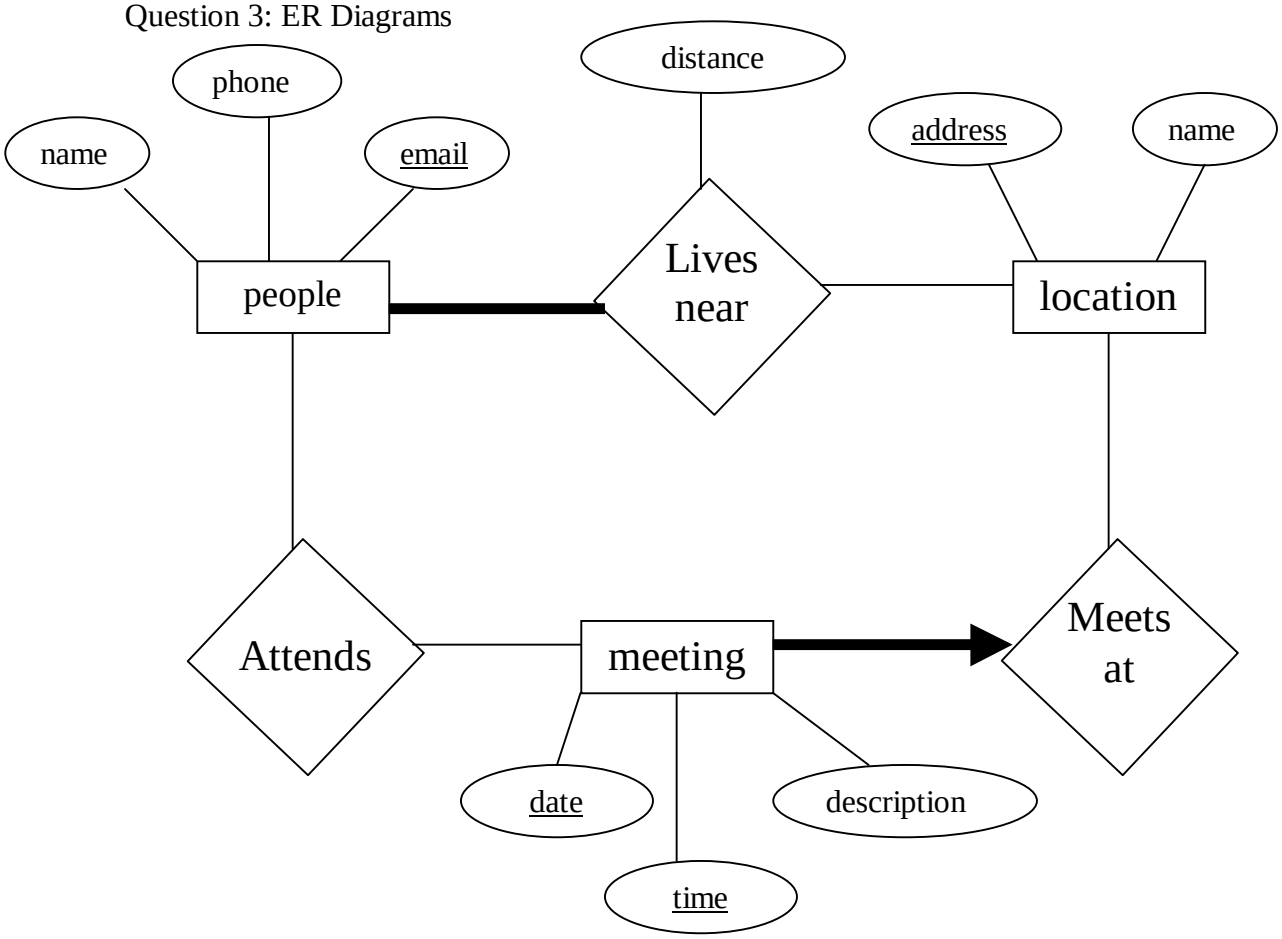
Solution 1:

```
SELECT S.filename, S.title
FROM Song S
WHERE NOT EXISTS( SELECT *
                  FROM User U1
                  WHERE NOT EXISTS ( SELECT *
                                     FROM Has H, User U2
                                     WHERE U2.speed = U1.speed
                                     AND H.name = U2.name
                                     AND S.filename = H.filename
```

Solution 2:

```
SELECT S.filename, S.title
FROM    Song S
WHERE NOT EXISTS (      SELECT U1.speed
                        FROM User U1
                        EXCEPT
                        SELECT U2.speed
                        FROM Has H, User U2
                        WHERE H.filename = S.filename
                        AND      H.name = U2.name
                  )
```

Question 3: ER Diagrams



Question 4: Join Algorithms

a) 5 points

[CARTS] + [CARTS] * [CONTENTS] = 1,000 + 1,000 * 5,000
= **5,001,000**

b)
[CARTS] + [CONTENTS] + ([CARTS] –1) * ([CONTENTS] – 999)
= 1,000 + 5,000 + (999 * 4,001)
= 6000 + 4,001,000 – 4,001 = **4,002,999**

c)

[CONTENTS] + ⌈ [CONTENTS] / 1,000 ⌉ * [CARTS]
= 5,000 + 5 * 1,000 = 10,000

d)

1. Hash Join, since √ [CARTS] fits in memory (2 passes)
2. Sort-Merge Join, since √ [CONTENTS] Does not fit in memory (3 passes)
3. Block Nested Loops Join, 20 passes of CONTENTS !